

Group 100 at YZV405E: Solving AdMIRE 2.0: Advancing Multimodal Idiomaticity Representation using Frozen Encoders, Dynamic Paraphrasing and Heuristic Ranking

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Abstract

Computational models fail to process Potentially Idiomatic Expressions (PIE), frequently interpreting them literally instead. In this project, we address the AdMIRE 2.0 image and text subtask, a multilingual benchmark requiring models to rank candidate images based on how accurately they reflect the intended meaning of a PIE within context. We propose a hybrid, three-step architecture that extends recent modular approaches in the literature. Rather than fine-tuning massive models, we combine frozen Vision-Language Encoders with dynamic idiom paraphrasing generated by a local, quantized Small Language Model. The final step relies on a traditional Logistic Regression combined with a deterministic ranking heuristic. Using this method, we aim to successfully address the exact-match weaknesses of generative ranking.

1 Introduction

Natural Language Processing systems traditionally rely on the principle of compositionality, where the meaning of a sentence is derived from its constituent words. However, figurative language and multiword expressions (MWEs) break this assumption. In this project, we focus on the AdMIRE 2.0 image and text subtask, which evaluates models on their ability to interpret PIEs using both text and images across 15 distinct languages. The primary goal is to order a set of five candidate images according to how well they capture the true semantic meaning of a context sentence featuring a multiword expression. Our project proposes a 3 step modular architecture that combines idiom paraphrasing with rule-based ranking to reduce literal visual bias.

2 Related Work

The technical focus on idioms has moved beyond basic text classification to prioritize the creation of deep semantic representations. (Zhai et al., 2023)

introduced SigLIP. By treating image-text alignment as a series of independent binary classification tasks, SigLIP achieves superior performance in zero-shot retrieval and multimodal grounding, particularly in large-scale and multilingual settings. (Li et al., 2023) introduced BLIP-2 which is a framework that efficiently bootstraps vision-language pretraining by bridging a frozen image encoder and a frozen LLM through a lightweight Querying Transformer (Q-Former). This architecture significantly reduces computational overhead while achieving state-of-the-art performance in zero-shot image-to-text generation and multimodal reasoning. To prevent models from anchoring to literal visual features, Hosseini-Kivanani (2026) employed an inline text rewriting strategy that physically replaced idiomatic expressions with non-figurative synonyms before feature extraction. While this approach improved zero-shot alignment in high-resource languages, the study demonstrated severe performance degradation in low-resource cross-lingual settings. Madabushi et al. (2022) established that models frequently fail to align idiomatic phrases within semantic text similarity tasks because purely compositional embeddings cannot capture figurative shifts. With the advent of multimodal architectures, foundational Vision-Language models such as CLIP (Radford et al., 2021) attempted to align text and image semantics. However, these dual-encoder models exhibit a severe "literal bias" that acts essentially as bag-of-words (Yuksekgonul et al., 2022). (Robertson and Zaragoza, 2009) provide the definitive framework for BM25, a probabilistic relevance model that remains the industry standard for keyword-based information retrieval. Umut and Senceylan demonstrated that separating sense prediction from final semantic ranking yields significant improvements in exact-match accuracy. Similarly, Cotiga and Nisioi successfully bridged literal and figurative meanings by generating auxiliary textual ex-

planations before utilizing an ensemble of models. Our approach follows this direction by combining frozen embeddings with explicit ranking rules, avoiding costly end-to-end training.

3 Proposed Methodology

Our architecture employs a 3-step pipeline combining frozen deep learning encoders with traditional deterministic logic.

3.1 Frozen Encoders and Dynamic Paraphrasing

Rather than fine-tuning a massive VLM, we utilize frozen encoders (SigLIP2 for vision, BGE-M3 for multilingual text). To overcome literal bias, we introduce an intermediate semantic translation step. We will utilize a local, quantized Small Language Model (e.g, Aya-23-8B or Phi-3.5) to dynamically generate straightforward, plain-language explanations that capture the figurative meaning of each idiom. By replacing the idiom with its explicit meaning prior to multimodal encoding, we aim to ensure the frozen VLM computes semantic alignment rather than literal word-to-object matching.

Zero-shot multilingual transfer suffers from tokenization fragmentation: low-resource languages such as Georgian, Uzbek and Igbo are poorly covered in multilingual encoder training data. This causes severe performance drops. To mitigate, we apply universal Test-Time Translation as a pre-processing step. Before any encoding occurs, all context sentences are automatically translated into English. Keeping all inputs in English ensures the encoder operates in its strongest latent space, preventing sub-word literal interpretations on fragmented tokenizations.

3.2 Traditional Logistic Sense Prediction

We chose Logistic Regression over a generative classifier for three reasons. First, training on frozen BGE-M3 embeddings produces embeddings that are compact and rich enough for the literal/idiomatic distinction to be linearly separable, making a linear model sufficient. Secondly, unlike generative models, Logistic Regression outputs a calibrated probability distribution rather than a hard label, which is a necessary condition for the confidence-gated bypass mechanism described in Section 3.3. Third, its inference cost is negligible and does not slow down the demo pipeline.

3.3 Category-Aware Heuristic Ranking

The final image ranking is executed via a hard-coded, deterministic heuristic based on the Logistic Regression output. If classified as Literal, the algorithm mathematically prioritizes images with high visual-literal feature overlap. If Idiomatic, the algorithm penalizes literal visual matches and prioritizes images whose visual content aligns with the paraphrased idiomatic meaning.

A critical vulnerability in any sequential pipeline is cascading errors: if the Logistic Regression misclassifies an idiomatic sentence as literal, the heuristic in Step 3 will actively penalize the correct images. To mitigate this, we aim to implement a Confidence-Gated Bypass. The heuristic ranking rules are only triggered if the classifier’s confidence exceeds a threshold of $P > 0.75$. If the classifier is uncertain, the system bypasses the deterministic rules entirely and falls back to the raw cosine similarity scores, which do not assume a fixed sentence type.

4 Experimental Setup and Feasibility

4.1 Dataset, Metrics

We will evaluate our system on the AdMIRe 2.0 image and text subtask dataset across the 15 required target languages (ZH, KA, EL, IG, KK, NO, PT-BR, PT-PT, RU, SR, SK, SL, ES-EC, TR, UZ). System performance will be measured strictly via top image accuracy and Discounted Cumulative Gain (DCG) utilizing the custom task-specific position weights. We will benchmark our pipeline against a vanilla zero-shot SigLIP2 baseline.

4.2 Limitations, Feasibility

By substituting computationally expensive generative ranking with frozen encoders and a quantized 8B-parameter SLM, our architecture is feasible.

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